

HASSAN HASHMI

linkedin/hassan-hashmi — github/KuroMaple — hassan-hashmi.com — hashmihassan47@gmail.com — (647) 270-5503

Technical Skills

Languages: C++, JavaScript(TypeScript), Python, Dart, C#, Kotlin, HTML, CSS, SQL

Technologies: React, Flutter, Django, Git, Claude, Vue, Unreal Engine

Experience

Software Developer

Jul. 2025 - Pres.

i3 Solutions Inc

Mississauga, ON

- Eliminated **730 annual database outages** and reduced client file upload **HTTP 500 errors to zero** by debugging end-to-end data flow across a production **Drupal/MySQL** full-stack application
- Modernized client-server system from **Vue 2/.NET 5** to **Vue 3/.NET 10**; cleared **46** dependency vulnerabilities
- Reduced user search page load times **30%** by optimizing inefficient **C# ORM operations**; simplified **MySQL** queries
- Shipped features for insurance platform serving **1M+ concurrent users** via **ASP.NET C#, React, and SQL**
- Resolved **30+** production defects by triaging and fixing issues across legacy **PHP and WordPress applications**
- Improved website visibility for product and business decisions by implementing **Google Analytics** across company site
- Mitigated XSS vulnerability in production by building **HTML sanitization service** in **C#** to validate user input

Software Developer

May 2023 - Aug. 2024

JANA Corporation

Aurora, ON

- Wrote **CRUD RESTful APIs** with **C# and T-SQL**; replaced third party services to save JANA **\$10,000** annually
- Delivered front-end features for a SaaS platform serving **25M+ households** using **React, TypeScript, and Redux**
- Built internal profit-sharing platform with **.NET and React** to automate financial calculations for **200+ employees**
- Optimized web form validation with **React Hook Form** and **Zod**; reduced database rollback instances by **75%**
- Wrote **50+** unit and integration tests using **React Testing Library and Vitest**, achieving **95%** code coverage
- Worked on a cross-functional team in an **Agile development** environment following **Scrum methodology**

Systems Programming - Instructional Assistant

Jan. 2023 - Apr. 2025

University of Waterloo

Waterloo, ON

- Taught **C++** to **1,200+** students by delivering tutorial presentations; covered core language features and best practices
- Guided students through building a C-like language compiler, from syntax parsing to binary code generation

Projects

Boids Algorithm Simulation | C++, SFML

- Built a high-performance **C++** application capable of simulating **5,000+** autonomous agents at **50 FPS** by implementing quadtree spatial partitioning for optimized neighbor searches
- Created a lightweight **SFML-based 2D rendering pipeline**; enforced render batching and update separation

Shinobi Bounce | Unreal Engine, C++

- Engineered scalable Unreal Engine gameplay framework in **C++**; leveraged **Object-Oriented Programming principles** to create reusable player ability systems, enemy AI logic, and real-time UI feedback systems
- Implemented **UMG menu system with Blueprint integration** for in-game music selection and player interactions

PokéView | React, TypeScript

- Crafted a Pokémon analytics dashboard supporting **650+** unique Pokémon with interactive search and data visualization
- Featured custom card flipping animation using **Framer Motion**, enhancing user interaction and visual appeal

Education

University of Waterloo

Sep. 2019 - Jun. 2025

Bachelors of Computer Science

- **Courses:** Data Structures & Algorithms(C++), Object-Oriented Programming (C++), Operating Systems (C)